Qatar-1b

R-1329

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_Qatar-1_210529` · created 2026-07-31T20:23:15+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459363.8406 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.018
Transit depth (Rp/Rs)^2	3.06 +/- 0.63 [%]
Orbital Inclination (inc)	83.07 +/- 0.81
Airmass coefficient 1 (a1)	1.233 +/- 0.02
Airmass coefficient 2 (a2)	-0.158 +/- 0.018
Scatter in the residuals of the lightcurve fit is	1.65 %
Best Comparison Star	None
Optimal Aperture	3.28
Optimal Annulus	9.68
Transit Duration (day)	0.0655 +/- 0.0057

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.175 ± 0.018 vs 0.1463 (deviation 1.3σ)
Duration fit vs published	0.0655 d vs 0.0692 d (deviation 0.52σ)
Mid-time O–C	5.2 min
Depth SNR	7.9
Residual scatter	1.65 %

Light curve

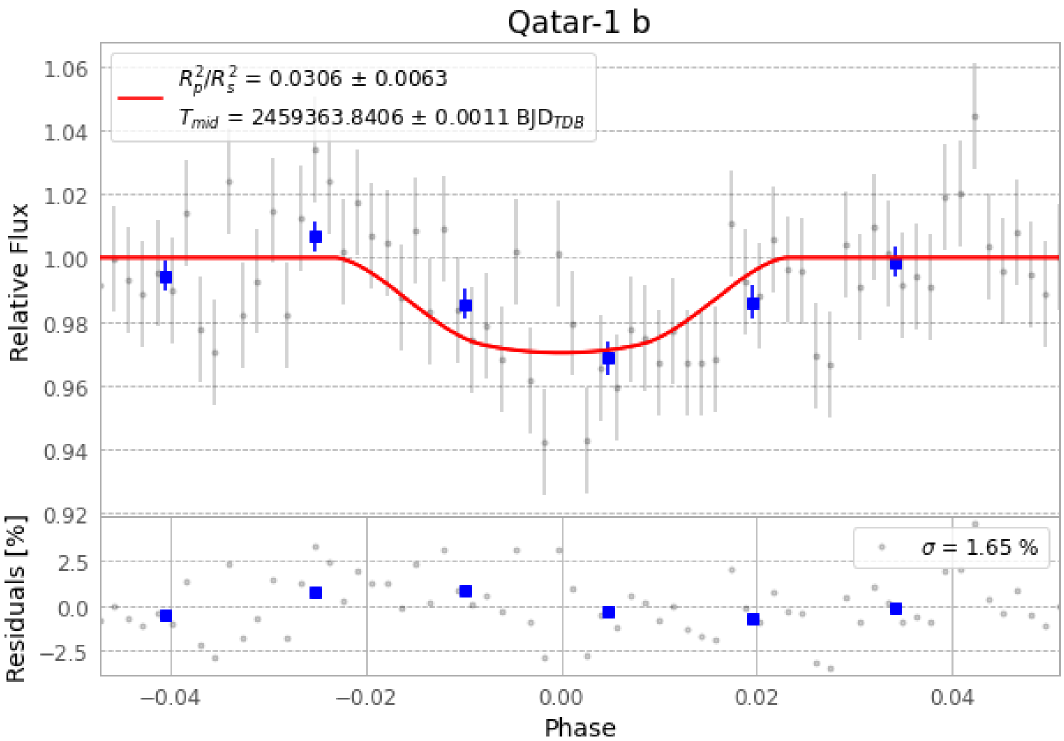

Figure 1 — FinalLightCurve_Qatar-1 b_2021-05-28.png (SHA-256 80140daaa8ef759a…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [318, 311], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 74e9b2d136901daa…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

68 FITS frames (45 MB), `Qatar-1210529063312.FITS` → `Qatar-1210529095412.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-210529.log (cedfdb4632af…), FinalLightCurve_Qatar-1 b_2021-05-28.png (80140daaa8ef…), FinalLightCurve_Qatar-1 b_2021-05-28.csv (e7afa917933b…)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)